Let $a_1,\dots,a_k$ be distinct positive integers with product P and sum $S<3\cdot 10^6$. Define

$$g=\gcd\left(\frac{P}{a_i}+1 : 1\le i\le k\right).$$

0.0.1 1. Basic congruences

For each i, $g\mid \frac{P}{a_i}+1$; hence $\frac{P}{a_i}\equiv -1 \pmod g$. Multiplying by a_i gives $P\equiv -a_i \pmod g$. Thus all a_i are congruent modulo g. Let r be the common residue, $0\le r<g$; then

$$a_i=r+g\,c_i \qquad (c_i \text{ distinct non-negative integers}).$$

If $r=0$ then each a_i is a multiple of g. Since $k\ge 3$, $\frac{P}{a_i}$ is a multiple of g, so $\frac{P}{a_i}+1\equiv 1 \pmod g$ and therefore $g=1$. Hence for any interesting case we have $r\ge 1$.

For any i,

$$\frac{P}{a_i}=\prod_{j\ne i}a_j\equiv r^{\,k-1} \pmod g.$$

Because $\frac{P}{a_i}\equiv -1 \pmod g$, we obtain

$$r^{\,k-1}\equiv -1 \pmod g.$$

0.0.2 2. The sum bound

The three smallest positive numbers in the residue class $r \pmod g$ are $r,\ r+g,\ r+2g$ (since $r\ge 1$). Their sum is $3r+3g$. The actual sum S is at least this sum, so

$$3r+3g<3\cdot 10^6 \quad\Longrightarrow\quad r+g<10^6.$$

0.0.3 3. The case $k\ge 4$ cannot beat the maximum

For $k\ge 4$ the smallest k numbers in the residue class are $r,\ r+g,\dots,\ r+(k-1)g$; their sum is

$$kr+g\cdot\frac{k(k-1)}{2}.$$

Because S is at least this sum,

$$kr+g\cdot\frac{k(k-1)}{2}<3\cdot 10^6.$$

For $k=4$ this gives $4r+6g<3\cdot 10^6$, i.e. $2r+3g<1.5\cdot 10^6$. Since $r\ge 1$, we obtain $3g<1.5\cdot 10^6-2$, so $g\le 499\,999$. For larger k the coefficient of g is even larger, so g is even smaller. Hence every configuration with $k \geq 4$ satisfies $g \leq 499\,999$. Therefore the overall maximum must be attained for $k = 3$.

0.0.4 4. The case $k = 3$

For $k = 3$ equation (1) becomes $r^2 \equiv -1 \pmod{g}$; hence

$$g \mid r^2 + 1.$$

Write $r^2 + 1 = g\,t$ with $t \in \mathbb{N}$. Together with (2) we have

$$r + \frac{r^2 + 1}{t} < 10^6.$$

We now maximise g under (3) and (4).

0.0.5 5. Maximising g

We consider the possible values of t.

- $t = 1$: $g = r^2 + 1$. Condition (4) becomes $r^2 + r + 1 < 10^6$. The largest integer r satisfying this is $r = 999$, giving $g = 999^2 + 1 = 998\,002$.
- $t = 2$: $g = \dfrac{r^2 + 1}{2}$. Condition (4) becomes $r + \dfrac{r^2 + 1}{2} < 10^6 \iff (r + 1)^2 < 2 \cdot 10^6$. Since $1414^2 = 2\,002\,225 > 2 \cdot 10^6$ and $1413^2 = 1\,996\,569 < 2 \cdot 10^6$, we have $r + 1 \leq 1414$, i.e. $r \leq 1413$. Moreover r must be odd (because $r^2 + 1$ is even). The largest odd $r \leq 1413$ is $r = 1413$. Then

  $$g = \frac{1413^2 + 1}{2} = \frac{1\,996\,569 + 1}{2} = 998\,285.$$

- $t \geq 3$: We show that $g \leq 998\,285$. Suppose, for contradiction, that $g \geq 998\,286$. From (2) we have $r < 10^6 - g \leq 1714$, so $r \leq 1713$. Then $r^2 + 1 \leq 1713^2 + 1 = 2\,934\,570$. Since $t \geq 3$,

  $$g = \frac{r^2 + 1}{t} \leq \frac{r^2 + 1}{3} \leq \frac{2\,934\,570}{3} = 978\,190.$$

  This contradicts $g \geq 998\,286$. Hence $g \leq 998\,285$ for every $t \geq 3$.

Thus the largest possible g for $k = 3$ is at most $998\,285$, and we already have a construction achieving this value.

0.0.6 6. Construction attaining $g = 998\,285$

Take $r = 1413$ and $g = 998\,285$. Set

$$a_1 = 1413, \qquad a_2 = 1413 + 998\,285 = 999\,698, \qquad a_3 = 1413 + 2 \cdot 998\,285 = 1\,997\,983.$$

These are distinct positive integers. Their sum is

$$1413 + 999\,698 + 1\,997\,983 = 2\,997\,094 < 3 \cdot 10^6.$$

Now compute:

$$\begin{aligned}
\frac{P}{a_1} + 1 &= a_2 a_3 + 1 = (r+g)(r+2g) + 1 = r^2 + 3rg + 2g^2 + 1 \\
&= (r^2+1) + g(3r+2g) = gt + g(3r+2g) = g(t + 3r + 2g), \\
\frac{P}{a_2} + 1 &= a_1 a_3 + 1 = r(r+2g) + 1 = r^2 + 2rg + 1 = (r^2+1) + 2rg = gt + 2rg = g(t+2r), \\
\frac{P}{a_3} + 1 &= a_1 a_2 + 1 = r(r+g) + 1 = r^2 + rg + 1 = (r^2+1) + rg = gt + rg = g(t+r).
\end{aligned}$$

Here $t = \dfrac{r^2+1}{g} = \dfrac{1413^2+1}{998\,285} = \dfrac{1\,996\,570}{998\,285} = 2$. Thus

$$\frac{P}{a_1} + 1 = g(2 + 3 \cdot 1413 + 2 \cdot 998\,285) = g \cdot 2\,000\,811, \qquad \frac{P}{a_2} + 1 = g \cdot 2 \cdot 1414 = g \cdot 2828, \qquad \frac{P}{a_3} + 1 = g \cdot (2 + 1413) = g \cdot 14$$

Since $\gcd(2828, 1415) = 1$ (because $2828 = 2 \cdot 1414$ and 1415 is odd and coprime to 1414), the three numbers have greatest common divisor exactly g. Hence $g = 998\,285$ is attainable.

0.0.7 7. Conclusion

The largest possible value of g over all choices of $k \geq 3$ and distinct positive integers a_i is

$$\boxed{998285}.$$